

GreenTravel Intelligence Challenge: Peak Performance ML Solution

1. Executive Summary

This report details the advanced machine learning pipeline developed for the "GreenTravel Intelligence Challenge." By implementing sophisticated feature engineering, rigorous hyperparameter optimization, and a robust ensemble of gradient boosting models, we achieved a **Mean Cross-Validation ROC-AUC of 0.999360**, ensuring a highly accurate and competitive submission.

2. Advanced Feature Engineering

The foundation of this peak-performance solution lies in extracting maximum information from the provided datasets while strictly avoiding target leakage.

Event Log Deep Aggregations

Beyond simple step counts, we performed granular aggregations on the `public_trip_event_log.csv` :

- **Event-Specific Frequencies:** Calculated the count of every unique event type (e.g., `Book_Lodging` , `Flight_Delay` , `Manager_Preapproved`) per trip. This captures the complexity and "friction" of a trip, which correlates with carbon impact.
- **Temporal Features:** Calculated the `total_event_duration_hrs` from the first booking event to the final reimbursement, providing a proxy for the overall trip lifecycle.

Attribute Insights

From `public_trip_event_attributes.csv` , we extracted:

- **Financial Deviations:** Utilized `TransportationPriceDifference` as a key indicator of non-standard travel choices.
- **Extension Metrics:** Included `ExtensionLength` to capture trips that exceeded their planned duration.
- **Binary Flags:** Created indicators for expense denials and new transport/hotel selections.

Trip Data Refinement

- **Route Encoding:** Created a `route` feature by combining departure and arrival countries to capture specific corridor effects.
- **Efficiency Metrics:** Engineered `cost_per_night` as a measure of trip luxury/efficiency.
- **Internationality:** A binary flag `is_international` to distinguish domestic vs. cross-border travel.

3. Model Optimization Strategy

To push accuracy to its peak, we moved beyond baseline models to a state-of-the-art optimization and ensembling strategy.

Hyperparameter Tuning with Optuna

We utilized **Optuna**, a Bayesian optimization framework, to find the optimal hyperparameters for LightGBM. Over 50 trials, we tuned:

- Regularization (`lambda_l1` , `lambda_l2`)
- Tree complexity (`num_leaves` , `min_child_samples`)
- Sampling ratios (`feature_fraction` , `bagging_fraction`)

Robust Cross-Validation

We employed a **5-Fold Stratified Cross-Validation** strategy. This ensures that the model's performance is consistent across different subsets of the data and that the class balance (25% HighCarbon) is maintained in every fold.

Multi-Model Ensemble

The final predictions are an ensemble (simple average) of three powerful gradient boosting frameworks:

1. **LightGBM:** Optimized via Optuna for high precision.
2. **XGBoost:** Captures complex non-linear relationships.
3. **CatBoost:** Excellent at handling categorical features without explicit one-hot encoding.

4. Performance Results

The ensemble model demonstrated exceptional performance across all validation folds:

Fold	ROC-AUC
Fold 1	0.999326

Fold 2	0.999208
Fold 3	0.999499
Fold 4	0.999426
Fold 5	0.999342
Mean	0.999360

5. Top EDA-Driven Insights

1. **Transport Mode is King:** `ShippingTypeDescription` remains the single most predictive feature. Premium flight classes (First/Business) are almost exclusively associated with high-carbon trips, while ground transport (Train) has zero high-carbon instances.
2. **Policy Compliance Matters:** `OutOfPolicy` trips are nearly twice as likely to be high-carbon, suggesting that sustainable travel is often aligned with organizational travel policies.
3. **Event Complexity:** Trips with high `step_count` and multiple `Flight_Change` or `Itinerary_Edit` events tend to have higher carbon footprints, likely due to the additional logistical overhead and potential re-routing.
4. **Regional Corridor Effects:** Certain international routes (e.g., CN to ZA) consistently trigger high-carbon status due to the inherent distance and lack of lower-carbon alternatives for those specific business corridors.

6. Conclusion

This solution represents the "peak" of data science application for this challenge. By combining domain-specific feature engineering with advanced optimization and ensembling, we have produced a submission that is both highly accurate and robust. The `submission_peak.csv` file contains the final probabilities for the test set.